

MATH 2790 Fall 2025

Homework 4

Due date: Wednesday, Sep 17

Instructions:

- (a) It is expected to show detailed work for every problem. You are allowed to have a discussion with peers on some challenging questions. But then you should work on your own to finish the assignment.
- (b) All questions in this assignment are from sections 1.12–2.3.

Problems:

1. Question (3) on page 78 of textbook under section 1.12.
2. Given that $a(t) = 1.04^t(1 - 0.05t)^{-1}$, use two methods to calculate δ_t and then calculate δ_5 .
3. Question (3) on page 111 of textbook under section 2.2. Round to 2 decimal places.
4. Question (1) on page 112 of textbook under section 2.3. Round to 2 decimal places.
5. Question (2) on page 112 of textbook under section 2.3. Round to 4 decimal places.
6. Question (3) on page 112 of textbook under section 2.3. Round to 2 decimal places.
7. Question (4) on page 112 of textbook under section 2.3. Use a financial calculator to get the final result. Round to 4 decimal places.
8. David can receive one of the following two payment streams:
 - 150 at time 0, 200 at time n and 400 at time $3n$.
 - 750 at time 10.

At an annual effective interest rate of i , the present values of the two streams are equal. Given $v^n = 0.75941$, determine i . Round to 4 decimal places.